HuangLab Nanopore sequencing workflow: amplicon (two-step or one-pot PCR) and genome

every stage ends with /lims-register · QR codes open the lab's protocol PDFs · LIMS stages 1 → 2 (strains) → 3a → 3b (amplicons) → 4a

Stage 1 — Specimen# / Culture#

Stage 2 — isolate + culture · Culture#

Stage 3a — extraction + Qubit · MolWork#

Stage 3b — PCR + gel + Qubit · pcr_reaction

Stage 4a — Nanopore experiment · Nanopore#

Specimen
soil / environmental / insect | fungal strain in culture

LIMS register `/lims-register` — Stage 1 → Specimen# (soil, env.) or Culture# (strain)

soil / environmental substrate: extract DNA directly (Route A, no culture)

isolate fungi (Route B/C strains)

Isolation and culturing
primary plate on MEA (Stage 2a) → subculture a single colony / hyphal tip on PDA
one Stage-2b row per isolate · preserve in vials
sequence from a pure culture only

Microbe isolation
environmental sample → vial

Culture media
MEA, PDA and other recipes

LIMS register `/lims-register` — Stage 2a plate → Stage 2b isolate → Culture# per strain

DNA extraction

DNA extraction
soil: SuperSoil, Zixpress 32 | fungal / HMW gDNA: BOMB v9
genome route: keep DNA high-molecular-weight

Soil DNA extraction
SuperSoil · Zixpress 32

Fungal DNA extraction
modified BOMB v9.0

quantify

Qubit Flex
concentration of the extract (dsDNA HS)



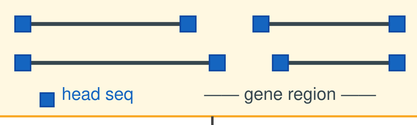
LIMS register `/lims-register` — Stage 3a → MolWork# (minted only after quantification)

choose a route

Route A · amplicon, two-step PCR

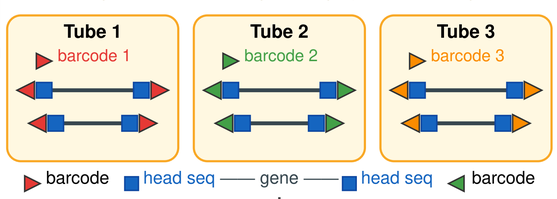
1st PCR (gene regions)

1st PCR products (per sample)



2nd PCR (barcoding)
gel check #1: good amplicons only

2nd PCR products: one tube per sample, one barcode per tube



2-step PCR protocol
mixtures + cycling

Route B · amplicon, one-pot PCR

Set up one-pot PCR

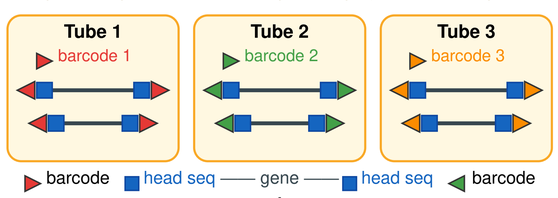
One reaction tube (per sample)

gene-region primers (head-tailed, 0.1 µM each) + **barcode primers** (barcode + head seq, 0.3 µM each)

50 µL: KAPA HiFi 2X + both primer sets + 3 µL template

One-pot PCR
phase 1: 25 cycles, 60–66 °C
phase 2: 15 cycles, 65 °C

One-pot PCR products: one tube per sample, one barcode per tube



One-pot PCR protocol
50 µL mix + 25/15 cycles

Route C · genome, no PCR

Stage 3a → Stage 4a directly (no 3b)

QC the gDNA

High-molecular-weight gDNA
integrity on gel / TapeStation; input per ONT table
(e.g. 10 kb avg → 200 fmol ≈ 1.3 µg)

Library construction

KAPA EvoPrep library + NBD114.96
end-prep → native barcode ligation (NB01–96)
pool barcoded samples → adapter ligation
0.5X reagent, PromethION · ~140 min

EvoPrep library prep
KAPA EvoPrep protocol

same barcoded molecule from either route



Gel check → Qubit Flex
confirm barcoded product, quantify → 10 ng/µL



LIMS register `/lims-register` — Stage 3b → pcr_reaction: method PCR_<Ta> / one_pot_<Ta>, conc., 2nd_pcr_ok, barcode

Pooling (equal molar)

Pool all barcoded samples
equal molar of every sample

Purification (bead clean-up)
PCR sample : bead = 1 : 1

Size selection 1X

Library construction

KAPA EvoPrep library
PCR-barcoded pool, ~200 fmol
end-prep → adapter ligation → clean-up
0.5X reagent, PromethION

EvoPrep library prep
KAPA EvoPrep protocol

LIMS register `/lims-register` — Stage 4a → Nanopore#: experiment + samples (PCR barcode or NB), flow cell, NanoAmp export

Load flow cell and sequence
PromethION R10.4.1 (FLO-PRO114M)

after the run

Flow cell wash and reuse
EXP-WSH004

Flow cell wash
EXP-WSH004 PromethION

Legend

- Head seq (universal tail on the gene-region primers)
- Barcode (sample-specific, on the barcode primers)
- Gene region (target amplicon; colour = different sample)

QR codes (left) open the lab's self-hosted protocol PDF for that step
left bands = LIMS stages; /lims-register at every stage; dashed = QC / info flow
Route C: native barcodes (NB) in the library; no PCR barcode, no Stage 3b
KAPA EvoPrep for every library; genome route adds NBD114.96 native barcodes